

Wood fusion for magmal comonads

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Wood's paper

- ▶ Richard Wood's 1978 paper "Coalgebras for closed comonads" considers a closed comonad G on a closed category $\mathcal{C}$. He provides necessary and sufficient conditions for the natural promonoidal structure on the category $\mathcal{C}^G$ of Eilenberg-Moore coalgebras for G to be closed (that is, to be representable by an internal hom). He also provides necessary and sufficient conditions for $\mathcal{C}^G$ to be closed in that way *and* for the underlying functor $\text{und} : \mathcal{C}^G \rightarrow \mathcal{C}$ to be strong closed (that is, to preserve the internal hom).

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- ▶ The process involves morphisms

$$w_{v,Z} := G[Y, Z] \xrightarrow{G_2^\ell} [GY, GZ] \xrightarrow{[v, 1]} [Y, GZ]$$

for $Y \xrightarrow{v} GY$ an object of $\mathcal{C}^G$ and Z an object of $\mathcal{C}$. These are what I will call *Wood fusion morphisms*.

Today's setting

- ▶ Like Wood, I will write for monoidal closed $\mathcal{C}$. Actually, it became clear that the story applies to more general categorical structures. In particular, we could work with algebras $\mathcal{C}$ for a club in the sense of Kelly (1972). It is about the functors providing the operations (such as binary tensor product), while the structural natural transformations (such as associativity constraints) and axioms thereon (such as the Mac Lane pentagon) carry over automatically to the constructions. For simplicity, I deal with *magma categories*; that is, categories $\mathcal{C}$ equipped with a binary functorial operation $\otimes : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$. Indeed, as is our custom, we (mostly implicitly) work with hom $\mathcal{V}$ -enriched categories throughout.

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- ▶ The unification of the monad and comonad cases is achieved by working with a magmal *procomonad* Γ on a magmal $\mathcal{C}$.

Magmal matters are as expected

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$$\begin{array}{ccc} \mathcal{C} \otimes \mathcal{C} & \xrightarrow{\quad \otimes \quad} & \mathcal{C} \\ S \otimes S \downarrow & \xRightarrow{\quad S_2 \quad} & \downarrow S \\ \mathcal{D} \otimes \mathcal{D} & \xrightarrow{\quad \otimes \quad} & \mathcal{D} \end{array}$$

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- ▶ A natural transformation $\theta : S \Rightarrow T : \mathcal{C} \rightarrow \mathcal{D}$ between magmal $\mathcal{V}$ -functors is *magmal* when it satisfies the equation $\theta_{C \otimes C'} \circ S_{2;C,C'} = T_{2;SC,SC'} \circ (\theta_C \otimes \theta_{C'})$ for all $C, C' \in \mathcal{C}$.

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- ▶ A comonad $G = (G, \varepsilon, \delta)$ is *magmal* when the functor $G : \mathcal{C} \rightarrow \mathcal{C}$, and the natural transformations $\varepsilon : G \Rightarrow 1_{\mathcal{C}}$ and $\delta : G \Rightarrow G \circ G$ are all magmal.

Cloaks, no daggers

Suppose $\mathcal{C}$ is a magmal category and $Y, Z \in \mathcal{C}$. We say Z is *left cloaked* by Y when there is an object $[Y, Z] \in \mathcal{C}$ and a morphism

$$\mathrm{ev}_Z^Y : [Y, Z] \otimes Y \rightarrow Z$$

such that the family

$$\mathcal{C}(X, [Y, Z]) \longrightarrow \mathcal{C}(X \otimes Y, Z) ,$$

natural in X obtained by the Yoneda Lemma, consists of isomorphisms. We say $[Y, Z]$ is the *left cloak of Z by Y* ; if these exist for all Z , we have the components

$$\mathrm{ev}_Z^Y : [Y, Z] \otimes Y \rightarrow Z \quad \text{and} \quad \mathrm{ve}_X^Y : X \rightarrow [Y, X \otimes Y]$$

of the counit and unit of an adjunction $- \otimes Y \dashv [Y, -]$. If left cloaks $[Y, Z]$ exist for all $Y, Z \in \mathcal{C}$, we say $\mathcal{C}$ is *left cloakal*.

Magmal Eilenberg-Moore coalgebras

Let G be a magmal comonad on a magmal category $\mathcal{C}$: we have natural families $\varepsilon_X : GX \rightarrow X$, $\delta_X : GX \rightarrow GGX$ and $G_{2;X,Y} : GX \otimes GY \rightarrow G(X \otimes Y)$. As for monoidal categories, the category $\mathcal{C}^G$ of Eilenberg-Moore G -coalgebras becomes magmal with

$$(X \xrightarrow{\xi} GX) \otimes (Y \xrightarrow{v} GY) = (X \otimes Y \xrightarrow{\xi \otimes v} GX \otimes GY \xrightarrow{G_{2;X,Y}} G(X \otimes Y)) .$$

This gives the construction of coalgebras for comonads in the 2-category $\text{Mag}(\mathcal{V}\text{-Cat})$ of magmal categories, magmal functors, and magmal natural transformations.

Cloaked coalgebras

Lemma (Wood technology)

Let G be a magmal comonad on a magmal $\mathcal{V}$ -category $\mathcal{C}$. Suppose cloaks $[(Y, v), (GZ, \delta_Z)]$ and $[(Y, v), (G^2Z, \delta_{GZ})]$ exist in $\mathcal{C}^G$ for the objects (Y, v) and (Z, ζ) of $\mathcal{C}^G$. Then the cloak $[(Y, v), (Z, \zeta)]$ exists if and only if the parallel pair

$$[(Y, v), (GZ, \delta_Z)] \begin{array}{c} \xrightarrow{[1, \delta_Z]} \\ \xrightarrow{[1, G\zeta]} \end{array} [(Y, v), (G^2Z, \delta_{GZ})] .$$

has an equalizer $(E, \kappa) \xrightarrow{k} [(Y, v), (GZ, \delta_Z)]$ in $\mathcal{C}^G$. If that holds, then

$$[(Y, v), (Z, \zeta)] = (E, \kappa)$$

and the counit $\text{ev}_{(Z, \zeta)}^{(Y, v)}$ is as in the commutative square (1) (PTO).

$$\begin{array}{ccc}
 (E, \kappa) \otimes (Y, v) & \xrightarrow{\text{ev}_{(Z, \zeta)}^{(Y, v)}} & (Z, \zeta) \\
 \downarrow k \otimes 1 & & \downarrow \zeta \\
 [(Y, v), (GZ, \delta_Z)] \otimes (Y, v) & \xrightarrow{\text{ev}_{(GZ, \delta_Z)}^{(Y, v)}} & (GZ, \delta_Z)
 \end{array} \tag{1}$$

Lemma (Wood)

Let G be a magmal comonad on a magmal $\mathcal{V}$ -category $\mathcal{C}$. Suppose the cloaks $[Y, Z]$ and $[GY, GZ]$ exist in $\mathcal{C}$. Then a cloak of the cofree object (GZ, δ_Z) by any object (Y, v) in $\mathcal{C}^G$ is given by the cofree object $(G[Y, Z], \delta_{[Y, Z]})$ with the composite

$$G[Y, Z] \otimes Y \xrightarrow{G_2^\ell \otimes \xi} [GY, GZ] \otimes GY \xrightarrow{\text{ev}_{GZ}^{GY}} GZ$$

as $\text{ev}_{(GZ, \delta_Z)}^{(Y, v)}$.

Wood fusion

Notice that, if the cloak $[Y, GZ]$ exists, then the evaluation in the last Lemma corresponds, under the universal property of cloaks, to:

$$w_{v,Z} := G[Y, Z] \xrightarrow{G_2^\ell} [GY, GZ] \xrightarrow{[v,1]} [Y, GZ] . \quad (2)$$

Definition

The $w_{v,Z}$ are called Wood G -fusion morphisms.

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We will be interested in when the Wood fusion morphisms are invertible. Dual to monad fusion, invertibility for an arbitrary G -coalgebra follows from invertibility for cofree G -coalgebras.

Proposition

For a G -coalgebra (Y, v) and any object Z , the Wood fusion morphism $w_{v,Z}$ is invertible if $w_{\delta_Y, Z}$ is invertible and $w_{\delta_{GY}, Z}$ is an epimorphism.

Proposition (Wood)

Let G be a magmal comonad on a magmal $\mathcal{V}$ -category $\mathcal{C}$. Suppose $(Y, v) \in \mathcal{C}^G$ is such that $[Y, Z]$ and $[GY, GZ]$ exist for all $Z \in \mathcal{C}$. The Wood G -fusion morphisms $w_{v,Z}$ are invertible for all Z if and only if $\text{und}_G : \mathcal{C}^G \rightarrow \mathcal{C}$ creates all cloaks by (Y, v) . In this case, for any $(Z, \zeta) \in \mathcal{C}^G$,

$$[(Y, v), (Z, \zeta)] \cong ([Y, Z], w_{v,Z}^{-1} \circ [1, \zeta]) .$$

Procomonads

- ▶ Let $\mathfrak{M} = \mathcal{V}\text{-Mod}$ be the bicategory of $\mathcal{V}$ -categories and $\mathcal{V}$ -modules: its homs are enriched in $\mathcal{V}\text{-Cat}$; we equate the hom $\mathfrak{M}(\mathcal{A}, \mathcal{B})$ with the $\mathcal{V}$ -functor $\mathcal{V}$ -category $[\mathcal{B}^{\text{op}} \otimes \mathcal{A}, \mathcal{V}]$.

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- ▶ A *procomonad* is a comonad in $\mathfrak{M}$ and so consists of a category $\mathcal{C}$, a module $\Gamma : \mathcal{C} \nrightarrow \mathcal{C}$, a natural transformation $\varepsilon : \Gamma \Rightarrow 1_{\mathcal{C}}$, and a natural transformation $\delta : \Gamma \Rightarrow \Gamma \circ \Gamma$ satisfying the coassociativity and counital conditions. We say $\Gamma = (\Gamma, \varepsilon, \delta)$ is a *procomonad on $\mathcal{C}$* .

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- ▶ For any category $\mathcal{A}$, we obtain a comonad $\mathfrak{M}(1_{\mathcal{A}}, \Gamma)$ on the category $\mathfrak{M}(\mathcal{A}, \mathcal{C})$. Then we have the category $\mathfrak{M}(\mathcal{A}, \mathcal{C})^{\mathfrak{M}(1_{\mathcal{A}}, \Gamma)}$ of Eilenberg-Moore $\mathfrak{M}(1_{\mathcal{A}}, \Gamma)$ -coalgebras.

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- ▶ In particular, when $\mathcal{A} = \mathcal{I}$, we obtain a comonad $\bar{\Gamma}$ on the presheaf category $[\mathcal{C}^{\text{op}}, \mathcal{V}]$ and its category $[\mathcal{C}^{\text{op}}, \mathcal{V}]^{\bar{\Gamma}}$ of Eilenberg-Moore $\bar{\Gamma}$ -coalgebras.

Thiébaud PhD (Dalhousie 1971)

The following definition agrees with the category $\mathcal{C}^\Gamma$ defined by Michel Thiébaud in the case $\mathcal{V} = \mathbf{Set}$.

Definition

The $\mathcal{V}$ -category of Γ -algebras in $\mathcal{C}$ is defined by the pullback in $\mathcal{V}\text{-Cat}$ of the underlying $\mathcal{V}$ -functor along the Yoneda embedding $\mathfrak{y}$.

$$\begin{array}{ccc} \mathcal{C}^\Gamma & \xrightarrow{\mathfrak{y}^\Gamma} & [\mathcal{C}^{\text{op}}, \mathcal{V}]^{\bar{\Gamma}} \\ \text{und} \downarrow & & \downarrow \text{und} \\ \mathcal{C} & \xrightarrow{\mathfrak{y}} & [\mathcal{C}^{\text{op}}, \mathcal{V}] \end{array}$$

More explicitly

Such a Γ -algebra consists of an object $C \in \mathcal{C}$ equipped with a *coaction* morphism $\gamma : I \rightarrow \Gamma(C, C)$, subject to the two axioms. We will write $\gamma_X : \mathcal{C}(C, X) \rightarrow \Gamma(C, X)$ for the natural family corresponding to γ under the Yoneda bijection. Similarly we have $\gamma^X : \mathcal{C}(X, C) \rightarrow \Gamma(X, C)$.

$$\begin{array}{ccc} I & & \\ \gamma \downarrow & \searrow 1_C & \\ \Gamma(C, C) & \xrightarrow{\varepsilon_{C,C}} & \mathcal{C}(C, C) \end{array}$$

$$\begin{array}{ccc} I & \xrightarrow{\gamma \otimes \gamma} & \Gamma(C, C) \otimes \Gamma(C, C) \\ \gamma \downarrow & & \downarrow \text{in}_C \\ \Gamma(C, C) & \xrightarrow{\delta_{C,C}} & \int^X \Gamma(X, C) \otimes \Gamma(C, X) \end{array}$$

We will call any $\mathcal{V}$ -functor into $\mathcal{C}$, isomorphic over $\mathcal{C}$ to $\text{und} : \mathcal{C}^\Gamma \rightarrow \mathcal{C}$, *Thiébaud algebraic over $\mathcal{C}$* .

The construction $\mathcal{C}^\Gamma$ includes the Eilenberg-Moore constructions for both monads and comonads.

Example

If $T = (T, \eta, \mu)$ is a $\mathcal{V}$ -monad on the $\mathcal{V}$ -category $\mathcal{C}$ and we take $\Gamma = T^*$ so that $\Gamma(X, Y) = \mathcal{C}(TX, Y)$ with counit ε and comultiplication δ induced by the unit η and multiplication μ then $\mathcal{C}^\Gamma \cong \mathcal{C}^T$, the $\mathcal{V}$ -category of T -algebras.

Example

If $G = (G, \varepsilon, \delta)$ is a $\mathcal{V}$ -comonad on the $\mathcal{V}$ -category $\mathcal{C}$ and we take $\Gamma = G_*$ so that $\Gamma(Y, Z) = \mathcal{C}(Y, GZ)$ with counit and comultiplication induced by those of G then $\mathcal{C}^\Gamma \cong \mathcal{C}^G$, the $\mathcal{V}$ -category of G -coalgebras.

The two main closure properties Thiébaud proved were that his algebraicity is closed under pullback and exponentiation. We now look at that.

Given a functor $W : \mathcal{D} \rightarrow \mathcal{C}$ and a procomonad Γ on $\mathcal{C}$, we have the procomonad $\Gamma_W = W^* \circ \Gamma \circ W_*$ on $\mathcal{D}$: *the lifting of Γ through W_* in $\mathfrak{M}$.*

Proposition

The following square is a pullback.

$$\begin{array}{ccc}
 \mathcal{D}^{\Gamma_W} & \xrightarrow{\quad} & \mathcal{C}^{\Gamma} \\
 \text{und} \downarrow & & \downarrow \text{und} \\
 \mathcal{D} & \xrightarrow{\quad W \quad} & \mathcal{C}
 \end{array}$$

Proof.

Using Yoneda, we deduce that $\Gamma_W(D', D) \cong \Gamma(WD', WD)$. □

Corollary

Thiébaud algebraicity is the closure under pullback of comonadicity.

Given categories $\mathcal{A}$ and $\mathcal{C}$, each procomonad Γ on $\mathcal{C}$ defines a procomonad $\Gamma^{\mathcal{A}}$ on $[\mathcal{A}, \mathcal{C}]$ thus:

$$\begin{array}{ccc}
 [\mathcal{A}, \mathcal{C}] & \xrightarrow{\Gamma^{\mathcal{A}}} & [\mathcal{A}, \mathcal{C}] \\
 [1_{\mathcal{A}}, \mathcal{I}]_* \downarrow & & \uparrow [1_{\mathcal{A}}, \mathcal{I}]^* \\
 [\mathcal{A}, [\mathcal{C}^{\text{op}}, \mathcal{V}]] & \xrightarrow{[1_{\mathcal{A}}, \bar{\Gamma}]_*} & [\mathcal{A}, [\mathcal{C}^{\text{op}}, \mathcal{V}]]
 \end{array}$$

It is the lifting of $[1_{\mathcal{A}}, \bar{\Gamma}]_*$ through $[1_{\mathcal{A}}, \mathcal{I}]_*$ and satisfies the simple formula:

$$\Gamma^{\mathcal{A}}(F', F) = \int_A \Gamma(F'A, FA) .$$

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 [1_{\mathcal{A}}, \mathcal{L}]_* \downarrow & & \uparrow [1_{\mathcal{A}}, \mathcal{L}]^* \\
 [\mathcal{A}, [\mathcal{C}^{\text{op}}, \mathcal{V}]] & \xrightarrow{[1_{\mathcal{A}}, \bar{\Gamma}]_*} & [\mathcal{A}, [\mathcal{C}^{\text{op}}, \mathcal{V}]]
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Proposition

There is an isomorphism $[\mathcal{A}, \mathcal{C}]^{\Gamma^{\mathcal{A}}} \cong [\mathcal{A}, \mathcal{C}^{\Gamma}]$ over $[\mathcal{A}, \mathcal{C}]$.

Proof.

Pullback and the Eilenberg-Moore construction of coalgebras are preserved by exponentiation $[\mathcal{A}, -]$ by $\mathcal{A}$ in $\mathcal{V}\text{-Cat}$. So we have the pullback

$$\begin{array}{ccc}
 [\mathcal{A}, \mathcal{C}^\Gamma] & \xrightarrow{\quad} & [\mathcal{A}, [\mathcal{C}^{\text{op}}, \mathcal{V}]]^{[1_{\mathcal{A}}, \bar{\Gamma}]} \\
 [1_{\mathcal{A}}, \text{und}] \downarrow & & \downarrow \text{und} \\
 [\mathcal{A}, \mathcal{C}] & \xrightarrow{[1_{\mathcal{A}}, \mathcal{J}]} & [\mathcal{A}, [\mathcal{C}^{\text{op}}, \mathcal{V}]] .
 \end{array}$$

The result now follows from the Proposition about pullback with $W = [1_{\mathcal{A}}, \mathcal{J}]$, the definition of $\Gamma^{\mathcal{A}}$, and the second Example a few slides back (the one about comonads). □

A module $M : \mathcal{C} \nrightarrow \mathcal{D}$ between magmal categories is called *magmal* when it is equipped with a natural family M_2 of morphisms

$$M_{2;C,C'}^{D,D'} : M(D, C) \otimes M(D', C') \rightarrow M(D \otimes D', C \otimes C') ;$$

such families are in bijection with natural families of morphisms

$$M_{2;C,C'}^{D,D'} : \int^{D,D'} M(D, C) \otimes M(D', C') \otimes \mathcal{D}(D'', D \otimes D') \rightarrow M(D'', C \otimes C') ;$$

and in bijection with natural families of morphisms

$$M_{2;C''}^{D,D'} : \int^{C,C'} \mathcal{C}(C \otimes C', C'') \otimes M(D, C) \otimes M(D', C') \rightarrow M(D \otimes D', C'') .$$

A module morphism $\alpha : M \Rightarrow N$ is *magmal* when

$$\alpha_{D \otimes D', C \otimes C'} \circ M_{2;C,C'}^{D,D'} = N_{2;C,C'}^{D,D'} \circ (\alpha_{D,C} \otimes \alpha_{D',C'}) .$$

A functor $S : \mathcal{C} \rightarrow \mathcal{D}$ is magmal if and only if the module S_* is.

Day convolution

If $\mathcal{C}$ is a small magmal category then the presheaf category $\hat{\mathcal{C}} = [\mathcal{C}^{\text{op}}, \mathcal{V}]$ has the *Day convolution magmal structure* $\star : \hat{\mathcal{C}} \otimes \hat{\mathcal{C}} \rightarrow \hat{\mathcal{C}}$ defined by

$$(F \star F')Z = \int^{X,Y} \mathcal{C}(Z, X \otimes Y) \otimes FX \otimes F'Y .$$

The Yoneda embedding $\mathfrak{y} : \mathcal{C} \rightarrow \hat{\mathcal{C}}$ is strong magmal and has the (bicategorical) universal property of the magmal small cocompletion of $\mathcal{C}$: for any small-cocomplete magmal $\mathcal{X}$, the category of [strong-]magmal functors $\mathcal{C} \rightarrow \mathcal{X}$ is equivalent (via left Kan extension along $\mathfrak{y}$) to the category of colimit-preserving [strong-]magmal functors $\hat{\mathcal{C}} \rightarrow \mathcal{X}$.

Magmal procomonads

A procomonad $\Gamma = (\Gamma, \varepsilon, \delta)$ on a magmal category $\mathcal{C}$ is *magmal* when $\Gamma, \varepsilon, \delta$ are all magmal. Then $\bar{\Gamma}$ is a magmal comonad on $\hat{\mathcal{C}}$. We can then lift to obtain a magmal structure on $\hat{\mathcal{C}}^{\bar{\Gamma}}$. This restricts along the fully faithful functor $\downarrow^{\Gamma} : \mathcal{C}^{\Gamma} \rightarrow \hat{\mathcal{C}}^{\bar{\Gamma}}$ to a magmal structure on $\mathcal{C}^{\Gamma}$ which is defined by

$$\begin{aligned} & \left(X, \downarrow \xrightarrow{\xi} \Gamma(X, X) \right) \otimes \left(Y, \downarrow \xrightarrow{v} \Gamma(Y, Y) \right) \\ & := \left(X \otimes Y, \downarrow \xrightarrow{\xi \otimes v} \Gamma(X, X) \otimes \Gamma(Y, Y) \xrightarrow{\Gamma_{2;XY}^{XY}} \Gamma(X \otimes Y, X \otimes Y) \right). \end{aligned}$$

Proposition (Day technology)

For any magmal $\mathcal{V}$ -category $\mathcal{C}$, the convolution magmal presheaf $\mathcal{V}$ -category $\hat{\mathcal{C}}$ is left cloakal. For $H, K \in \hat{\mathcal{C}}$, the left cloaking of K by H is given by

$$[H, K]U = \int_V [HV, K(U \otimes V)] \cong \hat{\mathcal{C}}(H, K(U \otimes -)) .$$

In particular, $[\mathfrak{y} Y, \mathfrak{y} Z]X \cong \mathcal{C}(X \otimes Y, Z)$ so that the Yoneda embedding $\mathfrak{y} : \mathcal{C} \rightarrow \hat{\mathcal{C}}$ preserves any left cloakings $\mathcal{C}$ admits.

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Definition

The fusion morphisms for magmal procomonad Γ are the Wood fusion morphisms for the magmal comonad $\bar{\Gamma}$ restricted to representables.

Let us make this definition more explicit.

Wood fusion for $\bar{\Gamma}$ is the composite

$$w_{\rho,K} := \bar{\Gamma}[H, K] \xrightarrow{\bar{\Gamma}_2^\ell} [\bar{\Gamma}H, \bar{\Gamma}K] \xrightarrow{[\rho, 1]} [H, \bar{\Gamma}K],$$

for $(H, \rho) \in \hat{\mathcal{C}}^{\bar{\Gamma}}$ and $K \in \mathcal{C}$. For $X, Z \in \mathcal{C}$ and $(Y, v) \in \mathcal{C}^{\Gamma}$, put $K = \downarrow Z$, $H = \downarrow Y$, and $\rho = v^-$, and write $w_{X,v,Z}$ for $(w_{\rho,K})_X$. Then

$$(\bar{\Gamma}[\downarrow Y, \downarrow Z])X \cong \int^U \mathcal{C}(U \otimes Y, Z) \otimes \Gamma(X, U), \quad [\downarrow Y, \bar{\Gamma} \downarrow Z]X = \Gamma(X \otimes Y, Z)$$

Proposition

Let Γ be a magmal procomonad on the magmal $\mathcal{V}$ -category $\mathcal{C}$. Suppose $(Y, I \xrightarrow{v} \Gamma(Y, Y)) \in \mathcal{C}^{\Gamma}$. Commutativity holds in diagram:

$$\begin{array}{ccc} \int^U \mathcal{C}(U \otimes Y, Z) \otimes \Gamma(X, U) & \xrightarrow{w_{XvZ}} & \Gamma(X \otimes Y, Z) \\ \int^U 11v \downarrow & & \uparrow \Gamma_{2;Z}^{XY} \\ \int^U \mathcal{C}(U \otimes Y, Z) \otimes \Gamma(X, U) \otimes \Gamma(Y, Y) & \xrightarrow{\int^U \text{in}_Y} & \int^{UV} \mathcal{C}(U \otimes V, Z) \otimes \Gamma(X, U) \otimes \Gamma(Y, V) \end{array}$$

Corollary

Suppose Z is left cloaked by Y in the magmal $\mathcal{V}$ -category $\mathcal{C}$ and $(Y, v) \in \mathcal{C}^\Gamma$. Then the fusion morphism $w_{X,v,Z}$ becomes the composite:

$$\begin{array}{ccc}
 \Gamma(X, [Y, Z]) & \xrightarrow{w_{X,v,Z}} & \Gamma(X \otimes Y, Z) \\
 1 \otimes v \downarrow & & \uparrow \Gamma(1, \text{ev}) \\
 \Gamma(X, [Y, Z]) \otimes \Gamma(Y, Y) & \xrightarrow{\Gamma_2} & \Gamma(X \otimes Y, [Y, Z] \otimes Y)
 \end{array}$$

Definition

We call Γ *Hopf at* (Y, v) when w_{XvZ} is invertible for all X, Z .

Lemma

Suppose $(Y, v) \in \mathcal{C}^\Gamma$ is such that $[Y, Z] \in \mathcal{C}$ exists for all $Z \in \mathcal{C}$. Then Γ is Hopf at (Y, v) if and only if $\bar{\Gamma}$ is Hopf at $\bowtie^\Gamma(Y, v)$.

Proof.

"If" is clear since w_{XvZ} is defined as a special case of Wood fusion for $\bar{\Gamma}$. Conversely, suppose

$w_{XvZ} : \Gamma(X, [Y, Z]) \cong \Gamma(X \otimes Y, Z)$ for all $X, Z \in \mathcal{C}$. For $K \in \hat{\mathcal{C}}$, one easily calculates that $[\bowtie Y, K] \cong K(- \otimes Y)$, so

$$\begin{aligned}\bar{\Gamma}[\bowtie Y, K] &\cong \int^U K(U \otimes Y) \otimes \Gamma(-, U) \\ &\cong \int^{UZ} \mathcal{C}(U \otimes Y, Z) \otimes KZ \otimes \Gamma(-, U) \cong \int^{UZ} \mathcal{C}(U, [Y, Z]) \otimes KZ \otimes \Gamma(-, U) \\ &\cong \int^Z KZ \otimes \Gamma(-, [Y, Z]) \cong \int^Z KZ \otimes \Gamma(- \otimes Y, Z) \\ &\cong \bar{\Gamma}(K)(- \otimes Y) \\ &\cong [\bowtie Y, \bar{\Gamma}(K)] . \quad \square\end{aligned}$$

Suppose Z is left cloaked by Y in the magmal $\mathcal{V}$ -category $\mathcal{C}$ and $(Y, v), (Z, \zeta) \in \mathcal{C}^\Gamma$. Suppose Γ is Hopf at (Y, v) . Then we have $([Y, Z], \omega) \in \mathcal{C}^\Gamma$ defined by commutativity of:

$$\begin{array}{ccc}
 I & \xrightarrow{\omega} & \Gamma([Y, Z], [Y, Z]) \\
 \zeta \downarrow & & \downarrow w_{[Y, Z]vZ} \\
 \Gamma(Z, Z) & \xrightarrow{\Gamma(\text{ev}, 1)} & \Gamma([Y, Z] \otimes Y, Z)
 \end{array} \tag{4}$$

Suppose Z is left cloaked by Y in the magmal $\mathcal{V}$ -category $\mathcal{C}$ and $(Y, v), (Z, \zeta) \in \mathcal{C}^\Gamma$. Suppose Γ is Hopf at (Y, v) . Then we have $([Y, Z], \omega) \in \mathcal{C}^\Gamma$ defined by commutativity of:

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 \Gamma(Z, Z) & \xrightarrow{\Gamma(\text{ev}, 1)} & \Gamma([Y, Z] \otimes Y, Z)
 \end{array} \tag{4}$$

Finally, we obtain our unification of opmagmal monad and magmal comonad as an application of the Wood work with $G = \bar{\Gamma}$.

Theorem

Let Γ be a magmal procomonad on a magmal $\mathcal{V}$ -category $\mathcal{C}$. Suppose $(Y, v) \in \mathcal{C}^\Gamma$ is such that $[Y, Z] \in \mathcal{C}$ exists for all $Z \in \mathcal{C}$. Then Γ is Hopf at (Y, v) if and only if $\text{und} : \mathcal{C}^\Gamma \rightarrow \mathcal{C}$ creates all cloaks by (Y, v) . In this case, for any $(Z, \zeta) \in \mathcal{C}^\Gamma$,

$$[(Y, v), (Z, \zeta)] \cong ([Y, Z], \omega),$$

where ω is defined by (4).

Lifting adjunctions and doctrinal adjunction

- ▶ Let $\mathbf{Mnd}\mathfrak{C}$ denote the bicategory of monads in a bicategory $\mathfrak{C}$.
- ▶ An object is a pair (A, s) consisting of an object $A \in \mathfrak{C}$ and a monoid s (called a *monad on A*) in the endomonoidal category $\mathfrak{C}(A, A)$. The unit and multiplication of s will be denoted by $\eta : 1_A \Rightarrow s$ and $\mu : s \circ s \Rightarrow s$.
- ▶ A morphism $(u, \phi) : (A, s) \rightarrow (B, t)$ (called a *monad morphism*) consists of a morphism $u : A \rightarrow B$ equipped with a 2-morphism

$$\begin{array}{ccc} A & \xrightarrow{s} & A \\ u \downarrow & \xRightarrow{\phi} & \downarrow u \\ B & \xrightarrow{t} & B \end{array} \quad (5)$$

in $\mathfrak{C}$ compatible with η, μ in the obvious way.

- ▶ A 2-morphism $\sigma : (u, \phi) \Rightarrow (v, \psi) : (A, s) \rightarrow (B, t)$ is a 2-morphism $\sigma : u \Rightarrow v$ in $\mathfrak{C}$ such that $\sigma s \circ \phi = \psi \circ t \sigma$.
- ▶ Composition is performed by pasting.
- ▶ There is a forgetful pseudofunctor

$$\mathrm{Mnd}\mathfrak{C} \rightarrow \mathfrak{C}, (A, s) \mapsto A. \quad (6)$$

- ▶ A morphism in $\mathrm{Mnd}^{\mathrm{op}}\mathfrak{C} = (\mathrm{Mnd}\mathfrak{C}^{\mathrm{op}})^{\mathrm{op}}$ is called a *monad opmorphism*: the 2-morphism in (5) is reversed. A morphism in $\mathrm{Mnd}^{\mathrm{co}}\mathfrak{C} = (\mathrm{Mnd}\mathfrak{C}^{\mathrm{co}})^{\mathrm{co}}$ is called a *comonad morphism*. A morphism in $\mathrm{Mnd}^{\mathrm{coop}}\mathfrak{C} = (\mathrm{Mnd}\mathfrak{C}^{\mathrm{coop}})^{\mathrm{coop}}$ is called a *comonad opmorphism*.

Example

- Let T be an opmagmal monad on the magmal $\mathcal{V}$ -category $\mathcal{C}$. For $(X, \alpha) \in \mathcal{C}^T$, T -fusion supplies a monad morphism

$$\begin{array}{ccc}
 \mathcal{C} & \xrightarrow{T} & \mathcal{C} \\
 - \otimes X \downarrow & \xRightarrow{\quad v_{-, \alpha} \quad} & \downarrow - \otimes X \\
 \mathcal{C} & \xrightarrow{T} & \mathcal{C} .
 \end{array}$$

- Let G be a magmal comonad on the magmal $\mathcal{V}$ -category $\mathcal{C}$. For $(Y, v) \in \mathcal{C}^G$, Wood fusion supplies a comonad opmorphism

$$\begin{array}{ccc}
 \mathcal{C} & \xrightarrow{G} & \mathcal{C} \\
 [Y, -] \downarrow & \xRightarrow{\quad w_{-, v} \quad} & \downarrow [Y, -] \\
 \mathcal{C} & \xrightarrow{G} & \mathcal{C} .
 \end{array}$$

Part of doctrinal adjunction is: if $(u, \phi) : (A, s) \rightarrow (B, t)$ is a monad morphism and $f \dashv u$ is an adjunction, then $(f, \hat{\phi}) : (B, t) \rightarrow (A, s)$ is a monad morphism where $\hat{\phi} : ft \Rightarrow sf$ is the mate of $\phi : tu \Rightarrow us$. The other part is obtained by examining adjunctions

$$(f, \theta) \dashv (u, \phi) : (A, s) \rightarrow (B, t) \quad (7)$$

in $\text{Mnd}\mathfrak{C}$. Since $\text{Mnd}\mathfrak{C} \rightarrow \mathfrak{C}$ preserve adjunctions, $f \dashv u$ in $\mathfrak{C}$, the counit $\alpha : fu \Rightarrow 1_A$ and the unit $\beta : 1 \Rightarrow uf$ must be 2-morphisms in $\text{Mnd}\mathfrak{C}$. Using only the first of these, a little diagram shows that θ has inverse the mate

$$\hat{\phi} : ft \xrightarrow{ft\beta} ftuf \xrightarrow{f\phi f} fuf \xrightarrow{\alpha sf} sf \quad (8)$$

of ϕ . Yet, if this mate of ϕ has an inverse at all, both α and β are 2-morphisms in $\text{Mnd}\mathfrak{C}$. This proves:

Proposition

A morphism $(u, \phi) : (A, s) \rightarrow (B, t)$ has a left adjoint in $\text{Mnd}\mathfrak{C}$ if and only if $u : A \rightarrow B$ has a left adjoint and the mate (8) of ϕ is invertible.

Adjunctions in the lax arrow bicategory

Let $\text{Fun}\mathfrak{C}$ denote the lax arrow bicategory of $\mathfrak{C}$. An object is a morphism $X \xrightarrow{x} A$ in $\mathfrak{C}$. A morphism $(u, v, \bar{u}) : x \rightarrow y$ is a diagram

$$\begin{array}{ccc} X & \xrightarrow{x} & A \\ \bar{u} \downarrow & \xRightarrow{v} & \downarrow u \\ Y & \xrightarrow{y} & B \end{array}$$

(9)

in $\mathfrak{C}$. A 2-morphism $(\sigma, \bar{\sigma}) : (u, v, \bar{u}) \Rightarrow (v, \omega, \bar{v}) : x \rightarrow y$ is a pair of 2-morphisms in $\mathfrak{C}$ satisfying $\sigma x \circ v = \omega \circ y \bar{\sigma}$. A morphism $(u, v, \bar{u})$ is called *strong* when v is invertible. Let $\text{sFun}\mathfrak{C}$ denote the sub-2-category of $\text{Fun}\mathfrak{C}$ obtained by restricting to the strong morphisms.

Proposition

A morphism $(u, v, \bar{u}) : x \rightarrow y$ has a left adjoint in $\text{Fun}\mathfrak{C}$ if and only if both $u : A \rightarrow B$ and $\bar{u} : X \rightarrow Y$ have left adjoints and the mate $\hat{v}$ of v is invertible. Any left adjoint in $\text{Fun}\mathfrak{C}$ is in $\text{sFun}\mathfrak{C}$. Any $(f, \tau, \bar{f})$ in $\text{sFun}\mathfrak{C}$ has a right adjoint in $\text{Fun}\mathfrak{C}$ if and only if f and $\bar{f}$ have right adjoints in $\mathfrak{C}$.

Eilenberg-Moore

Suppose $\mathfrak{C}$ admits the construction of algebras in the bicategorical sense: for each monad (A, s) , there is an Eilenberg-Moore s -algebra $x_s : A^s \rightarrow A$ with action $\xi_s : sx_s \rightarrow x_s$ inducing equivalences $\mathfrak{C}(X, A^s) \simeq \mathfrak{C}(X, A)^{\mathfrak{C}(1_X, s)}$. Then we have a pseudofunctor $\text{Mnd}\mathfrak{C} \xrightarrow{\text{EM}} \text{sFun}\mathfrak{C}$ defined as follows. For each monad (A, s) , we put $\text{EM}(A, s) = x_s$. For each monad morphism $(A, s) \xrightarrow{(u, \phi)} (B, t)$, we have a t -algebra $tux_s \xrightarrow{vx_s} usx_s \xrightarrow{u\xi_s} ux_s$, so there exist (uniquely up to isomorphism) a morphism $\bar{u} : A^s \rightarrow A^t$ and an isomorphism $x_t \bar{u} \xrightarrow{v} ux_s$ such that v becomes a t -algebra isomorphism from $(x_t \bar{u}, \xi_t \bar{u})$ to the previous t -algebra. We put $\text{EM}(u, \phi) = (u, v, \bar{u}) : x_s \rightarrow x_t$; it is a strong morphism. For a 2-morphism $\sigma : (u, \phi) \Rightarrow (v, \psi) : (A, s) \rightarrow (B, t)$, there is a unique 2-morphism $\bar{\sigma} : \bar{u} \Rightarrow \bar{v}$ such that $(\sigma, \bar{\sigma}) : (u, v, \bar{u}) \Rightarrow (v, \omega, \bar{v}) : x_s \rightarrow x_t$ is a 2-morphism in $\text{Fun}\mathfrak{C}$; we put $\text{EM}\sigma = (\sigma, \bar{\sigma})$.

Here is a restatement of an observation of Appelgate (1965).

Proposition

The pseudofunctor $\mathbf{Mnd}\mathfrak{C} \xrightarrow{\text{EM}} \mathbf{sFun}\mathfrak{C}$ is an equivalence on homcategories.

Corollary

A monad morphism $(u, \phi) : (A, s) \rightarrow (B, t)$ has a left adjoint in $\mathbf{Mnd}\mathfrak{C}$ if and only if $\text{EM}(u, \phi)$ has a left adjoint in $\mathbf{Fun}\mathfrak{C}$.

Corollary

A monad morphism $(u, \phi) : (A, s) \rightarrow (B, t)$ has a right adjoint in $\mathbf{Mnd}\mathfrak{C}$ if and only if $\text{EM}(u, \phi)$ has a right adjoint $(r, \rho, \bar{r})$ in $\mathbf{sFun}\mathfrak{C}$. In particular, if u has a right adjoint r and ϕ is invertible then $\bar{u}$ has a right adjoint $\bar{r}$ with $x_s \bar{r} \cong r x_t$.

Rue des Remarques

Remark

In general the $\mathcal{V}$ -functor $\text{und} : \mathcal{C}^\Gamma \rightarrow \mathcal{C}$ has neither left nor right adjoint. The opmagmal monad and magmal comonad examples are where it does.

Remark

*A recent paper with an unlikely-to-be-relevant title:
“Everybody knows what a normal gabi-algebra is”
by Johannes Berger, Paolo Saracco and Joost Vercruysse,
<https://arxiv.org/abs/2308.09449> (2023) 40pp.
does seem relevant. They look at skew-monoidal closedness for
Eilenberg-Moore categories for monads that do not start out being
opmonoidal. They have structure which is in the reverse direction to
fusion morphisms.*

Thank you one and all!

$$M \xrightarrow[\Delta]{\text{CoACT}} MQ$$